

Kuvvet Toplamları: Tarihi, Yapısı ve Uygulamaları

Doruk Üstündağ

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- 1 Preliminaries
- 2 History and Mathematical Structure of Power Sums
- 3 References

Definition of Power Sums

Definition (Power Sums)

Let k, n be any natural numbers. (We assume that $0 \in \mathbb{N}$)

$$S_k(n) = \sum_{i=1}^n i^k = 1^k + 2^k + 3^k + \cdots + (n-1)^k + n^k$$

Examples

$$S_0(n) = 1^0 + 2^0 + 3^0 + \cdots + n^0 = n$$

$$S_1(n) = 1^1 + 2^1 + 3^1 + \cdots + n^1 = \frac{n(n+1)}{2}$$

$$S_2(n) = 1^2 + 2^2 + 3^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

$$S_3(n) = 1^3 + 2^3 + 3^3 + \cdots + n^3 = \frac{n^2(n+1)^2}{4}$$

Pythagorean School

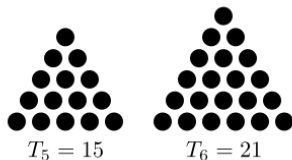
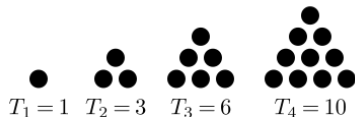


Michele Tedesco, *A Pythagorean School Invaded by Sybarites* (1887), Guildhall Art Gallery, London.

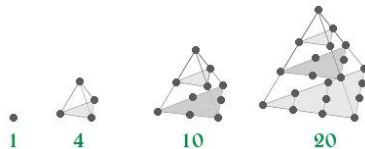
Geometric Idea

- Figurate Numbers

Geometric Idea



Triangular Numbers



Tetrahedral Numbers

First Two Figurate Numbers $\implies$ Power Sums

- $T_n = S_1(n)$
- $Te(n) = \sum_{i=0}^n S_1(i)$

Nicomachus of Gerasa



Nicomachus of Gerasa (60-120 AD)

- Pythagorean
- Introduction to Arithmetic
- Nicomachus' Theorem

Nicomachus' Theorem

Theorem (Nicomachus)

The sum of the first n cubed natural numbers is equal to the square of the sum of the first n natural numbers.

- $1^3 + 2^3 + 3^3 + \cdots + n^3 = (1 + 2 + 3 + \cdots + n)^2$
- $S_3(n) = (S_1(n))^2$

Allison's Result (1961)

Allison [1] has proved that the identity (in n)

$$\left(\sum_{r=1}^n r^k \right)^p = \left(\sum_{r=1}^n r^m \right)^q \quad (p > q)$$

holds only in the familiar case: $S_3(n) = (S_1(n))^2$



Pierre de Fermat (1601-1665)

- Recurrence relation for power sums based on figurate numbers in 1636



Blaise Pascal (1623-1662)

- Formula with "Binomial Expansion" in 1654
- The Arithmetic Triangle
- Pascal's Identity

Pascal's Identity to Recursion

Using Pascal's Identity to Find Recursion of Power Sums

Power sums $S_k(n) = 1^k + 2^k + \cdots + n^k$ can be expressed in terms of these coefficients because:

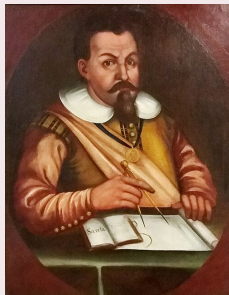
$$(i+1)^{m+1} - i^{m+1} = \sum_{j=0}^m \binom{m+1}{j} i^j.$$

Summing this identity from $i = 0$ to $n - 1$ yields a recurrence for power sums:

$$(n+1)^{m+1} - 1 = \sum_{j=0}^m \binom{m+1}{j} S_j(n),$$

which rearranges to:

$$S_m(n) = \frac{1}{m+1} \left[(n+1)^{m+1} - 1 - \sum_{j=0}^{m-1} \binom{m+1}{j} S_j(n) \right].$$



Johann Faulhaber (1580-1635)

- *Academia Algebrae* [5]
- Generalizations (Faulhaber's Polynomials)
- Up to S_{17} in 1631
- Human "computer" → "Rechenmeister"

Faulhaber's Polynomials

Faulhaber's Polynomials

Let $a = S_1(n)$. Faulhaber saw that all sums of odd powers can be written in the base of the first sum of powers:

$$\sum_{k=1}^n k^5 = \frac{4a^3 - a^2}{3}$$

$$\sum_{k=1}^n k^7 = \frac{6a^4 - 4a^3 + a^2}{3}$$

$$\sum_{k=1}^n k^9 = \frac{16a^5 - 20a^4 + 12a^3 - 3a^2}{5}$$

$$\sum_{k=1}^n k^{11} = \frac{16a^6 - 32a^5 + 34a^4 - 20a^3 + 5a^2}{3}$$

An idea of generalization

Faulhaber found a form for the power sum polynomial:

$$\sum n^r \text{ (} r \text{ odd)} = \text{a polynomial in } n(n+1)$$

$$\sum n^r \text{ (} r \text{ even)} = (2n+1) \times (\text{a polynomial in } n(n+1))$$

Jacob Bernoulli



Jacob Bernoulli (1655-1705)

- *Ars Conjectandi* [3]
- First mention of Faulhaber
- "Summae Potestatum"
- No attempt at generalization
- Bernoulli's Formula (Posthumous)

Summae Potestatum

... Atque si porrò ad altiores gradatim potestates pergere, levique negotio sequentem adornare laterculum licet :

Summae Potestatum

$$\begin{aligned} f n &= \frac{1}{2} n n + \frac{1}{2} n \\ f n n &= \frac{1}{3} n^3 + \frac{1}{2} n n + \frac{1}{6} n \\ f n^3 &= \frac{1}{4} n^4 + \frac{1}{2} n^3 + \frac{1}{4} n n \\ f n^4 &= \frac{1}{5} n^5 + \frac{1}{2} n^4 + \frac{1}{3} n^3 - \frac{1}{30} n n \\ f n^5 &= \frac{1}{6} n^6 + \frac{1}{2} n^5 + \frac{5}{12} n^4 - \frac{1}{12} n n \\ f n^6 &= \frac{1}{7} n^7 + \frac{1}{2} n^6 + \frac{1}{2} n^5 - \frac{1}{6} n^3 + \frac{1}{42} n n \\ f n^7 &= \frac{1}{8} n^8 + \frac{1}{2} n^7 + \frac{7}{12} n^6 - \frac{7}{24} n^4 + \frac{1}{12} n n \\ f n^8 &= \frac{1}{9} n^9 + \frac{1}{2} n^8 + \frac{2}{3} n^7 - \frac{7}{15} n^5 + \frac{2}{5} n^3 - \frac{1}{35} n \\ f n^9 &= \frac{1}{10} n^{10} + \frac{1}{2} n^9 + \frac{3}{4} n^8 - \frac{7}{10} n^6 + \frac{1}{4} n^4 - \frac{1}{12} n n \\ f n^{10} &= \frac{1}{11} n^{11} + \frac{1}{2} n^{10} + \frac{5}{6} n^9 - 1 n^7 + 1 n^5 - \frac{1}{2} n^3 + \frac{5}{66} n \end{aligned}$$

Quin imò qui legem progressionis inibi attentius enspexerit, eundem etiam continuare poterit absque his ratiociniorum ambabimus : Sumtā enim c pro potestatis cujuslibet exponente, fit summa omnium n^c seu

$$\begin{aligned} \int n^c &= \frac{1}{c+1} n^{c+1} + \frac{1}{2} n^c + \frac{c}{2} A n^{c-1} + \frac{c \cdot c-1 \cdot c-2}{2 \cdot 3 \cdot 4} B n^{c-3} \\ &+ \frac{c \cdot c-1 \cdot c-2 \cdot c-3 \cdot c-4}{2 \cdot 3 \cdot 4 \cdot 5 \cdot 6} C n^{c-5} \\ &+ \frac{c \cdot c-1 \cdot c-2 \cdot c-3 \cdot c-4 \cdot c-5 \cdot c-6}{2 \cdot 3 \cdot 4 \cdot 5 \cdot 6 \cdot 7 \cdot 8} D n^{c-7} \dots \& \text{ ita deinceps,} \end{aligned}$$

exponentem potestatis ipsius n continuè minuendo binario, quosque perveniat ad n vel nn. Literae capitales A, B, C, D & c. ordine denotant coefficientes ultimorum terminorum pro $f n n$, $f n^4$, $f n^6$, $f n^8$, & c. nempe

$$A = \frac{1}{6}, B = -\frac{1}{30}, C = \frac{1}{42}, D = -\frac{1}{30}.$$

Summae Potestatum (Sums of Powers) (Posthumous-1713)

- Coefficient of n^2 in $\sum n^9$ is wrong! ($-\frac{1}{2} \rightarrow -\frac{3}{20}$ corrected)

Bernoulli's Formula (Faulhaber's Formula)

The Well-known Formula

$$S_k(n) = 1^k + 2^k + 3^k + \cdots + n^k = \frac{1}{k+1} \sum_{j=0}^k \binom{k+1}{j} B_j n^{k+1-j}$$

Some discussion on the topic

- Faulhaber's attempt at formulation
- According to Knuth [7], the first proof of the formula was given by Jacobi.
- What are B_j 's?



Leonhard Euler (1707-1783)

- Calculus of finite differences
- "Bernoulli" Numbers

Bernoulli Numbers

Bernoulli Numbers

We are going to use $B_1^+ = \frac{1}{2}$,

$$\begin{array}{lll} B_0 = 1, & B_1 = +\frac{1}{2}, & B_2 = \frac{1}{6}, \\ B_3 = 0, & B_4 = -\frac{1}{30}, & B_5 = 0, \\ B_6 = \frac{1}{42}, & B_7 = 0, & B_8 = -\frac{1}{30}, \\ B_9 = 0, & B_{10} = \frac{5}{66}. \end{array}$$

A little note from A. W. F. Edwards

"Poor Faulhaber might reasonably have felt a little aggrieved at this..." [4]

Bernoulli Numbers

Explicit Form

All Bernoulli Numbers can be given in the form:

$$B_m^+ = \sum_{k=0}^m \frac{1}{k+1} \sum_{j=0}^k \binom{k}{j} (-1)^j (j+1)^m.$$

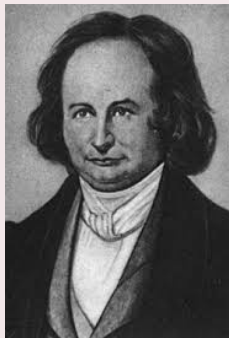
Generating Function Form

Bernoulli Numbers (B_m^+) can be given by the following generating function:

$$\frac{te^t}{e^t - 1} = \frac{t}{1 - e^{-t}} = \frac{t}{2} \left(\coth \frac{t}{2} + 1 \right) = \sum_{m=0}^{\infty} \frac{B_m^+ t^m}{m!}.$$

Relation to Riemann Zeta

$$B_k = -k\zeta(1-k)$$



Carl Gustav Jacob Jacobi (1804-1851)

- Euler-Maclaurin Summation Formula
- Rigorous Proof



Anthony William Fairbank Edwards (1935)

- Sums of powers of integers: A little of the history [4]
- Faulhaber's Memoir
- Matrix Form of Power Sum Polynomials

Sketch of the A. W. F. Edwards' Matrix Form

First five power sums

$$\sum_{k=1}^n k^0 = 1n$$

$$\sum_{k=1}^n k^1 = \frac{1}{2}n + \frac{1}{2}n^2$$

$$\sum_{k=1}^n k^2 = \frac{1}{6}n + \frac{1}{2}n^2 + \frac{1}{3}n^3$$

$$\sum_{k=1}^n k^3 = 0n + \frac{1}{4}n^2 + \frac{1}{2}n^3 + \frac{1}{4}n^4$$

$$\sum_{k=1}^n k^4 = -\frac{1}{30}n + 0n^2 + \frac{1}{3}n^3 + \frac{1}{2}n^4 + \frac{1}{5}n^5$$

Matrix Form of Power Sum Polynomials

Interpreting polynomials as products of matrices

$$\begin{pmatrix} \sum k^0 \\ \sum k^1 \\ \sum k^2 \\ \sum k^3 \\ \sum k^4 \end{pmatrix} = G_5 \begin{pmatrix} n \\ n^2 \\ n^3 \\ n^4 \\ n^5 \end{pmatrix}$$

G_5

$$G_5 = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 \\ \frac{1}{6} & \frac{1}{2} & \frac{1}{3} & 0 & 0 \\ 0 & \frac{1}{4} & \frac{1}{2} & \frac{1}{4} & 0 \\ -\frac{1}{30} & 0 & \frac{1}{3} & \frac{1}{2} & \frac{1}{5} \end{pmatrix}$$

Matrix Form of Power Sum Polynomials

$$G_5^{-1} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ -1 & 2 & 0 & 0 & 0 \\ 1 & -3 & 3 & 0 & 0 \\ -1 & 4 & -6 & 4 & 0 \\ 1 & -5 & 10 & -10 & 5 \end{pmatrix} = \bar{A}_5$$

seems more familiar than the G_5 matrix.

Getting closer to Pascal Matrix

Let A_5 be the matrix which is obtained by changing $a_{i,j}$ of $\bar{A}_5$ into $(-1)^{i+j}a_{i,j}$.

$$A_5 = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 1 & 2 & 0 & 0 & 0 \\ 1 & 3 & 3 & 0 & 0 \\ 1 & 4 & 6 & 4 & 0 \\ 1 & 5 & 10 & 10 & 5 \end{pmatrix}$$

Matrix Form of Power Sum Polynomials

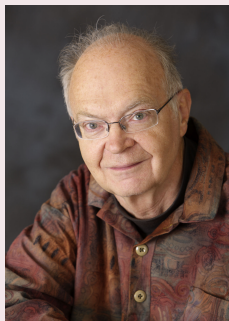
Converse of the Idea

Let $\bar{G}_5$ be the matrix which can be obtained by changing $g_{i,j}$ of G_5 into $(-1)^{i+j}g_{i,j}$. It is trivial to see that:

$$A_5^{-1} = \bar{G}_5$$

At general

For any natural number m , $G_m^{-1} = \bar{A}_m$ and $\bar{G}_m^{-1} = A_m$. So we can get any power sum polynomial inductively.



Donald Ervin Knuth (1938)

- Johann Faulhaber and sums of powers [7]
- Computation-based Approach
- Faulhaber's Cryptomath (!)
- Comment on the correctness of the Faulhaber's Results

Arthur Benjamin

- Two Quick Combinatorial Proofs [2]
- Gave a "bijective" proof of Nicomachus' Theorem

George Mackiw

- A Combinatorial Approach to Sums of Integer Powers [8]
- First idea of combinatorial interpretation

Idea of Binomial Basis

We define the *binomial polynomials* (b_n) in $\mathbb{Q}[z]$ for $n \in \mathbb{N}$ as follows:

$$b_0 = 1,$$

$$b_n = b_n(z) = \frac{1}{n!} \prod_{i \in [n]} z - i, \text{ where } n > 0.$$

Note that each b_i is of degree i .

Integer-valued Polynomials

$$\text{Int}(\mathbb{Z}) = \{f(z) \in \mathbb{Q}[z] : f(\mathbb{Z}) \subseteq \mathbb{Z}\}$$

Theorem 1 (Göksu, Ü.)

The polynomials $(b_i)_{i=0}^{\infty}$ (see Definition (3)) form a basis for $\mathbb{Q}[z]$ as a $\mathbb{Q}$ -vector space; for any polynomial $f \in \mathbb{Q}[z]$ of degree n , we have

$$f = \sum_{i \in [n+1]} (\Delta^i f(0)) b_i,$$

where $(\Delta^i f(0))$ is the evaluation of the polynomial $\Delta^i f$ (see Definition 1) at 0. Furthermore, $\text{Int}(\mathbb{Z})$ is a free $\mathbb{Z}$ -module generated by the polynomials (b_i) ; a polynomial $f \in \mathbb{Q}[z]$ of degree n is integer-valued if and only if the forward differences $\Delta^0 f(0), \dots, \Delta^n f(0)$ are all integers.

Set of implications that leads to power sums

- Inclusion-Exclusion Principle $\rightarrow$ Binomial Transform
- Binomial Transform $\rightarrow$ Theorem 1 [6]
- Theorem 1 $\rightarrow$ Pure Combinatorics to Algebra
- Image Factorization $\rightarrow$ Power Sums

Formula Born From Pure Combinatorics (Göksu, Ü.)

For any natural number $n > 0$, we have

$$S_n = \sum_{i=1}^{n+1} \left\{ \begin{matrix} n+1 \\ i \end{matrix} \right\} (i-1)! b_i.$$

What To Do With Combinatorial Interpretation

Divisibility

- Combinatorial interpretation gives us a rule for integer divisibility $S_3 \mid S_n$.

Computation

- Again, combinatorial interpretation gives us the formula given before. And this formula is now the best algorithm (time complexity-wise). By creating an $O(n^2)$ algorithm, we beat the $O(n^3)$ matrix form algorithm of Wang et al.[9].

Definition ($\mathcal{S}_k$)

Let's start by defining the set $\mathcal{S}_k$:

$$\mathcal{S}_k = \{n \in \mathbb{N} : S_k \mid S_n, n \geq k\} \text{ for given natural number } k \geq 1.$$

Examples of $\mathcal{S}_k$

$$\mathcal{S}_1 = \{\text{Set of all odd numbers}\}$$

$$\mathcal{S}_3 = \text{Open Problem}$$

Conjecture over $\mathcal{S}_3$

Conjecture (Göksu, Ü.)

We have

$$\mathcal{S}_3 = \bigcup_{i=0}^{\infty} C_{p_0, \dots, p_i}.$$

It is apparent that the union on the right contains $\mathcal{S}_3$. Yet for every $n \in [1000]$, either $S_3 \mid S_n$, or there exists some prime $p < n - 2$ for which $S_3(p - 1) \nmid S_n(p - 1)$ and $S_3(p) \nmid S_n(p)$. In fact, in all the aforementioned cases, the prime in question is the first positive integer at which S_3 fails to divide S_n .

Examples With SageMath

For the sake of giving an example for a better understanding of the form of $\mathcal{S}_3$, we have listed the elements of $\mathcal{S}_3$ less than 1000 with the help of SageMath:

{3, 15, 27, 39, 63, 75, 87, 99, 123, 135, 147, 159, 183, 195, 207, 219, 255, 267, 279, 285, 303, 315, 327, 339, 363, 387, 399, 405, 423, 435, 447, 459, 483, 495, 519, 543, 555, 567, 579, 603, 615, 627, 663, 675, 687, 699, 723, 735, 747, 759, 795, 807, 819, 825, 843, 855, 867, 879, 915, 927, 939, 963, 975, 999}.

To generate this subset of $\mathcal{S}_3$, it suffices to eliminate the failures at the following odd primes:

$$\mathcal{P} = \{3, 5, 11, 17, 23, 29, 47, 53, 59, 83, 173, 179, 263, 293\}.$$

That is to say, $\mathcal{S}_3 \cap [1000] = C_{\mathcal{P}} \cap [1000]$. It is worth noting that $\mathcal{P}$ is *not* the set of all odd primes in $[294]$. It is currently unclear whether every odd prime p eventually has $S_3(p-1) \nmid S_n(p-1)$ for some n while $S_3(x) \mid S_n(x)$ for every $x \in [p-1]$.

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THANK YOU ALL FOR LISTENING

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